

Software Requirements Specification

GOV-01: Government Scheme Finder (Citizen-Facing Retrieval System)

Document Version:
1.0.0

Status: Approved
Baseline

Target Audience: Developers /
Evaluators

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1. INTRODUCTION

1.1 Purpose

This document specifies the software requirements for the **Government Scheme Finder (GOV-01)** system. The application serves as an intelligent, citizen-facing discovery platform that simplifies how citizens identify, evaluate, and apply for government welfare schemes. By capturing basic personal attributes and requirements, the system retrieves relevant schemes, calculates match scores, and presents eligibility details in an accessible manner.

1.2 Scope

The solution covers the end-to-end process of citizen profiling, natural-language or questionnaire-driven requirement capturing, dataset query execution, hybrid semantic/rule-based matching logic, match score generation, and detailed transparency profiling ("Why this scheme matches you"). Out of scope for this version are direct online application submissions, document verification, and integration with government portal SSO APIs.

1.3 Definitions and Acronyms

- **SRS:** Software Requirements Specification
- **RAG:** Retrieval-Augmented Generation
- **NLP:** Natural Language Processing
- **Hybrid Matcher:** A retrieval engine combining strict Boolean filtering (rule-based) with vector semantic embedding comparisons.
- **Match Score:** A percentage rating (0–100%) indicating how closely a citizen's profile and needs align with a scheme's criteria.

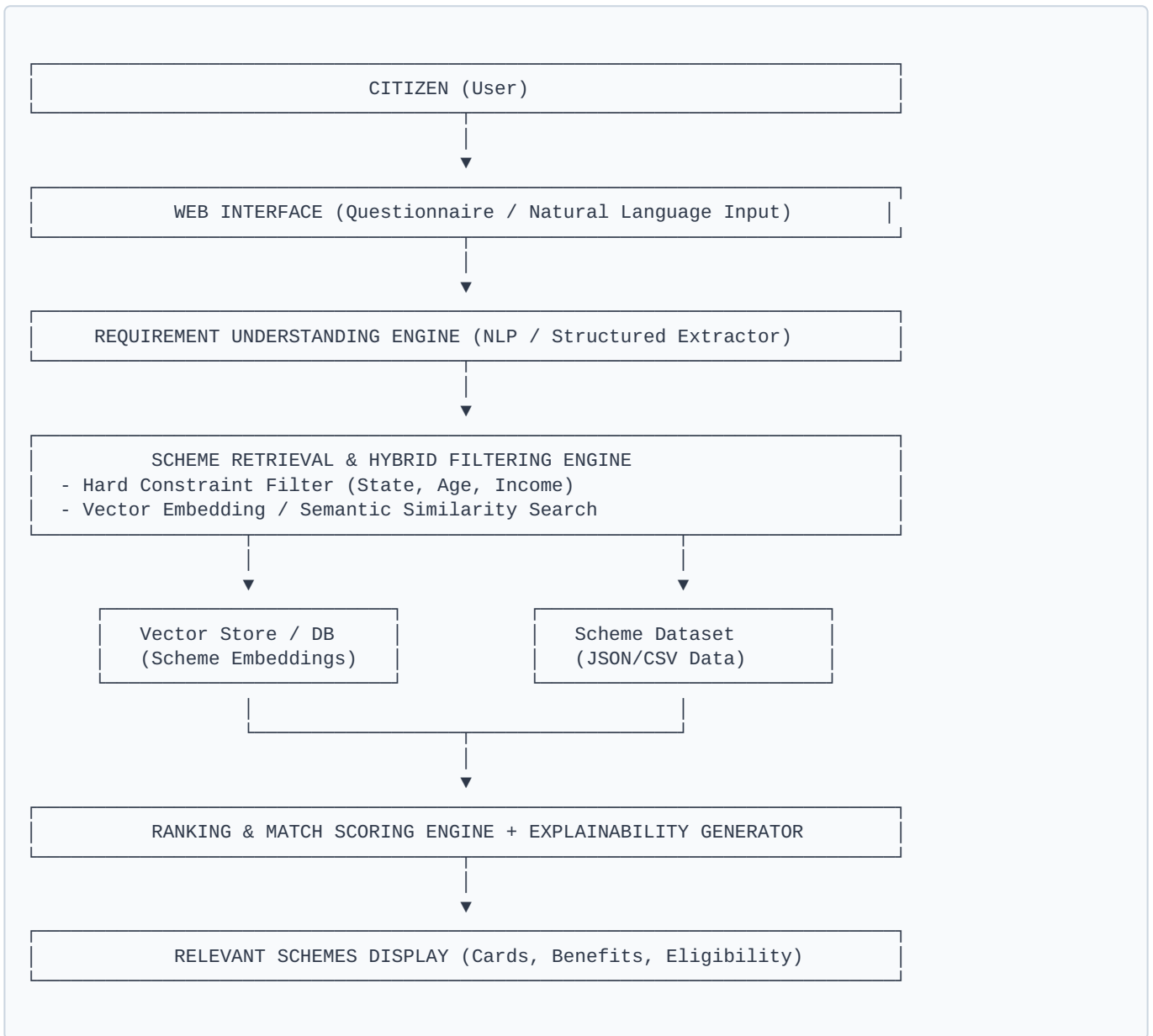
2. SYSTEM OVERVIEW & ARCHITECTURE

2.1 User Classes and Characteristics

- **General Citizens:** Primary users with varying technical literacy and regional language preferences. Requires clean, simple UI with zero jargon.
- **Nodal Officers / System Admins:** Secondary users responsible for uploading and maintaining the structured scheme dataset.

2.2 High-Level Architecture

The system utilizes a modern decoupled architecture where natural language input and structured questionnaire entries are processed through an AI-powered requirement understanding engine, matched via a hybrid retrieval pipeline, and returned with explainable criteria.



3. FUNCTIONAL REQUIREMENTS

3.1 Module 1: Citizen Profile & Requirement Capture

Req ID	Feature Name	Description	Priority
FR-1.1	Guided Questionnaire	Capture user fields: Age, State, Annual Income, Occupation, Education Level, Rural/Urban, Disability Status, and Primary Needs (e.g., Education, Farming, Healthcare).	High
FR-1.2	Natural Language Input	Allow citizens to type prompt-style queries (e.g., <i>"I am a college student from Tamil Nadu needing financial help"</i>). Extract key parameters via NLP.	High
FR-1.3	Form Validation	Validate ranges (Age: 0-120, Income ≥ 0) and state selection before proceeding to search.	Medium

3.2 Module 2: Scheme Retrieval & Matching Logic

Req ID	Feature Name	Description	Priority
FR-2.1	Deterministic Filtering	Filter out schemes where absolute criteria fail (e.g., scheme is state-specific to Kerala, but citizen resides in Tamil Nadu).	High
FR-2.2	Semantic Vector Search	Convert unstructured user requirements into vector embeddings and compare against embedded scheme summaries.	High
FR-2.3	Match Score Calculation	Compute an aggregated match percentage (e.g., 95% Highly Relevant) based on age suitability, income upper-bounds, and requirement alignment.	High
FR-2.4	Explainable Logic	Generate clear "Why you see this" explanations highlighting matched and partially matched attributes.	High

3.3 Module 3: UI Output & Scheme Display

Req ID	Feature Name	Description	Priority
FR-3.1	Scheme Card Display	Render results using intuitive visual cards containing Category, Match Rating, Benefits, and Basic Eligibility bullet points.	High
FR-3.2	Detailed Scheme Modal	Provide a "View Details" view displaying required application documents, official links, and contact channels.	Medium

Sample Result Card Specification



Post-Matric Scholarship Scheme for SC/ST

95% Match

Category: Education & Literacy | **State:** Tamil Nadu

Key Benefit: Full tuition fee reimbursement and annual maintenance allowance of up to ₹12,000.

Basic Eligibility:

- Age: 18–25 years
- Income Limit: Below ₹2,50,000 / year
- Target Group: Enrolled in higher education institutions

Why this matches: Your age (21) and annual income (₹2.0L) fall within the threshold for Tamil Nadu residents.

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance & Scalability

- **Response Time:** Search query results and scheme recommendations must return within < 1.5 seconds under normal load.
- **Dataset Scale:** Capable of indexing and searching over 10,000 scheme records without UI degradation.

4.2 Usability & Accessibility

- **Mobile-First Responsive Design:** The UI must render seamlessly across low-end smartphones and desktop browsers.
- **Simplicity:** Form fields must use visual icons and minimal text input to accommodate users with basic digital literacy.

4.3 Data Security & Privacy

- **Stateless Profile Handling:** Personal profile inputs (e.g., income, age) are processed in memory to fetch schemes and are not stored permanently without explicit consent.

5. SCHEME DATASET SCHEMA

The application relies on a structured schema for stored scheme records:

Field Name	Data Type	Sample Value	Description
scheme_id	String / UUID	SCH-TN-ED-004	Unique identifier for the scheme
scheme_name	String	Tamil Nadu Chief Minister Fellowship	Official title of the scheme
category	Enum / String	Education / Agriculture / Housing	Functional sector
eligible_states	List[String]	["Tamil Nadu"] or ["All India"]	State applicability
max_income	Float	250000.00	Upper income limit in INR
min_age / max_age	Integer	18 / 25	Age limits
benefits	String / Text	Monthly stipend of ₹20,000	Summary of benefits provided

6. EXECUTION & SUCCESS CRITERIA

The solution successfully satisfies problem statement **GOV-01** when a citizen can enter their basic demographics and natural language requirements, receive ranked schemes matching their profile within seconds, and understand the transparent matching criteria provided with each scheme card.